

Variable and model selection

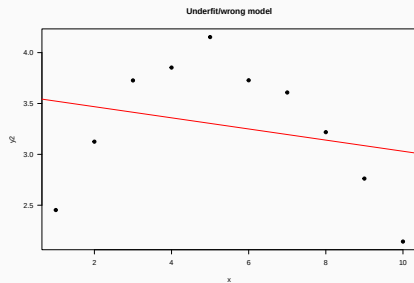
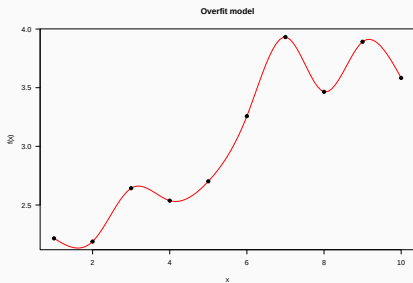
Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

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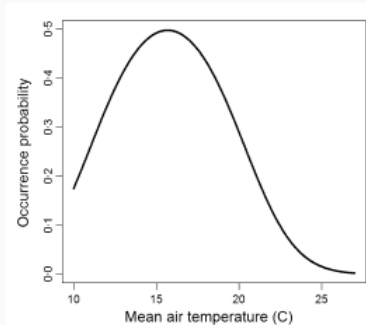
- On one hand, we want to **maximise fit**.
- On the other hand, we want to **avoid overfitting** and overly complex models.

Overfitting and balanced model complexity

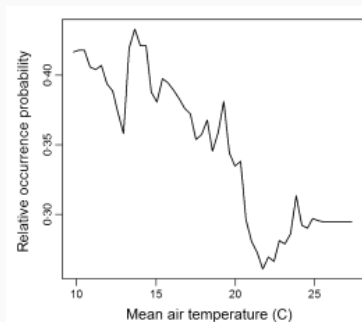


Overfitting and balanced model complexity

GLMM



Random forests



Wenger & Olden (2012)

Overfitted models will work badly on new data



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- All these methods have flaws!

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- AICc recommended with 'small' sample sizes ($n/p < 40$). But see [Richards 2005](#)

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- They estimate *average* out-of-sample prediction error. But errors can differ substantially within dataset.
- Sometimes better models rank poorly (e.g. see [Gelman et al. 2013](#)). Combine with **thorough model checks**.

So which variables should enter
my model?

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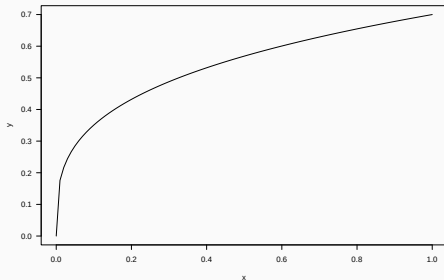
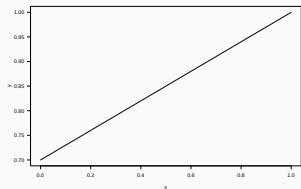
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 - Measurement error can seriously complicate things (Biggs et al 2009; Freckleton 2011)
- For predictors with large effects, **consider interactions**.

Think about the shape of relationships

$$y \sim x + z$$

Really? Not everything has to be linear! Actually, it often is not.

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Removing predictors

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- This includes **stepAIC** (e.g. Dahlgren 2010; Burnham et al 2011; Hegyi & Garamszegi 2011).

- Testing bivariate relationships before building multivariable model

Heinze & Dunkler 2016

Other common bad practices

- Testing bivariate relationships before building multivariable model
- Removing non-significant predictors

Heinze & Dunkler 2016

- Always **keep 'core' predictors** (based on previous knowledge)

Heinze et al 2018

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Removing predictors?

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- If performing variable selection, always **assess stability** (bootstrap, etc)

Heinze et al 2018

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5. Always report effect sizes